Innovation as prospective-but-not-retrospective surprise: Kullback–Leibler novelty scores on 2.0 million USPTO trademark filings

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Abstract

We operationalize firm-level innovation by applying Simon DeDeo's prospective/retrospective surprise framework to the goods-and-services descriptions of every U.S. trademark filed since 1985 across all 45 NICE international classes (16.5 million filings, 5.7 million unique registrants). Two classes are presented in detail throughout the paper: NICE Class 9 (Software & Electronics) and NICE Class 32 (Beer & Soft Drinks). The remaining 43 industries are summarized in Section 8. For each filing at calendar year t we compute the Kullback–Leibler divergence of its term distribution from a 5-year look-behind reference (*prospective* surprise) and from a 5-year look-ahead reference (*retrospective* surprise). The contrast $\Delta KL = KL_{\text{pros}} - KL_{\text{retr}}$ is positive when a filing was unusual at the time but normal in hindsight — the empirical signature of a firm that introduced a category later imitated — and negative when the past predicts the filing better than the future does.

The score recovers the risk–reward asymmetry that the literature on pioneers and fast-followers predicts. In a generalized linear logit on 1,146,415 Class-9 filings, a one-standard-deviation increase in ΔKL *lowers* the probability of reaching registration by 0.042 in log-odds (SE = 0.002) but *raises* the probability of being a currently-live mark in 2026 by 0.038 (SE = 0.002). Linking the trademark scores to SEC EDGAR financials for 1,597 matched publicly-traded firms gives the same pattern at the firm-year level: gross margin rises with ΔKL at +3.2 percentage points per σ (SE 0.6 pp). The decomposition shows that the gain comes from firms whose filings the future ends up resembling, not from firms whose filings stand alone. A separate within-firm variance regression confirms the symmetric prediction: firms whose recent filings score higher on ΔKL have measurably more variable gross margins, the empirical fingerprint of a portfolio that includes some pioneers who succeed and some who do not.

1 Background

Innovation is a central object of empirical economic and management research, but the construct is hard to measure. The dominant quantitative approaches are R&D expenditure (an input proxy that rewards spend rather than novelty), patent counts and patent citations (which require strategic legal filing, and which Trajtenberg, Hagedoorn, and others have shown to be at best a noisy proxy), and the market value premium of equity over book value (which conflates innovation with every other investor perception). Survey-based scales such as the Community Innovation Survey and Gatignon et al.'s instrument capture richer constructs but suffer from common-method bias and limited sample size. The lack of agreement on what to measure is itself a finding: across measures, pairwise correlations are typically $\rho \approx 0.3$.

We propose using trademark filings as a complementary novelty channel. Trademarks have several useful properties that patents do not: applicants file a single, brief, machine-readable goods-and-services description that is sharply incentivized to be exact (broad enough to protect future use, narrow enough to clear examination); the data are released into the public domain by the USPTO without lag; and the universe of filers is broader than the universe of patentees, including most consumer-facing firms. The cost is that trademarks measure novelty in the goods-and-services lexicon, not in the underlying technology — a brand that solves an old problem with a new identity (e.g. Liquid Death's canned water) is largely invisible.

Our broader motivation is to ask whether innovative firms are rewarded or punished. Two pieces of folk wisdom are in tension. The first, conveyed in the phrase "you can tell the pioneers by the arrows in their backs," is that firms that depart from convention bear higher search and adoption costs, get out-competed by fast-followers, and underperform on average. The second, expressed in the older saying "build a better mousetrap and the world will beat a path to your door," is that meaningful novelty attracts customers and rewards its inventor with durable advantage. Both can be true simultaneously: novelty might raise variance — some pioneers fail spectacularly, others enjoy durable category leadership. This paper offers an empirical test of that joint pattern using the full USPTO trademark backfile.

2 Data

We download the entire USPTO Trademark Annual Applications backfile (product TRTYRAP, $\sim$12.4 GB across 91 ZIP parts as of April 2026) through the Open Data Portal API. Records are streamed: each ZIP is downloaded, parsed in memory with `lxml.iterparse`, filtered to the target NICE international class, and written incrementally to Parquet, after which the raw ZIP is deleted. From each `<case-file>` record we retain the serial number, filing date, registration date, status code, mark identification, NICE class list, US class list, registrant name (the owner at the time of filing), and goods-and-services text (case-file statements with type-code prefix `GS`). Mark-description statements (prefix `D`) are excluded because they describe the visual mark, not the business.

All 45 NICE international classes are processed; in this paper we work through NICE Class 9 (Software & Electronics) and NICE Class 32 (Beer & Soft Drinks) in narrative detail before reporting the cross-industry comparison in Section 8. Class 9 was chosen as the canonical software class with high-volume long-tailed structure; Class 32 was chosen for its tractable size and well-known innovation cycles (microbrew, energy drinks, hard seltzer, cannabis beverages). Owner names are taken verbatim from the USPTO XML; entity resolution to canonical firm identifiers is described in Section 7. Per-class summary statistics for these two classes are reported in Appendix A; the analogous table for all 45 classes is in Section 8.

For firm-level outcome data we use the SEC EDGAR Financial Statement Data Sets (FSDS), 64 quarterly XBRL releases from 2009Q1 through 2024Q4. The FSDS covers 13,127 unique CIKs after filtering to consolidated 10-K income-statement and balance-sheet tags. We extract revenue, cost of revenue, gross profit, R&D expense, operating income, and net income.

3 Method

For each filing i at year t with goods-and-services text w_i we construct a term-frequency distribution P_i over a fixed dictionary V of unigram and bigram tokens. The dictionary is built once per NICE class with a custom analyzer that splits each description at semicolons before tokenizing, so bigrams cannot cross goods/services-item boundaries. We retain only unigrams and bigrams that appear in at least 50 distinct filings (documents) and in no more than 50% of all filings, on the view that very rare tokens are typos and very common tokens are USPTO boilerplate.

For each calendar year t we form two reference distributions by aggregating term frequencies across neighboring years:

$$Q_t^{\text{pros}}(v) \propto \alpha + \sum_{j:\text{year}(j)\in[t-W,t)} \text{count}(v,j), \tag{1}$$

$$Q_t^{\text{retr}}(v) \propto \alpha + \sum_{j:\text{year}(j)\in(t,t+W]} \text{count}(v,j), \tag{2}$$

with window $W = 5$ years (look-behind for prospective, look-ahead for retrospective) and Laplace smoothing $\alpha = 1.0$. The prospective and retrospective surprise of filing i are the support-restricted KL divergences

$$KL_i^{\text{pros}} = \sum_{v:P_i(v)>0} P_i(v)\,\log\frac{P_i(v)}{Q_t^{\text{pros}}(v)}, \quad KL_i^{\text{retr}} = \sum_{v:P_i(v)>0} P_i(v)\,\log\frac{P_i(v)}{Q_t^{\text{retr}}(v)}.$$

The directional innovation score is $\Delta KL_i = KL_i^{\text{pros}} - KL_i^{\text{retr}}$. We use a term-frequency multinomial as the v0 baseline; an LDA topic-model replication is planned but not reported here.

Identical-text filings are signal. Trademark practice rewards templated language. Many breweries file the boilerplate token `Beer`; many app developers file `downloadable software`. We do not deduplicate identical text: filings that use the same words *should* receive the same score, because deviations from the boilerplate are themselves the signal.

Burn-in caveat. The method requires a five-year window of prior filings (to compute prospective surprise) and a five-year window of subsequent filings (to compute retrospective surprise). The first means filings from the very beginning of the digitised USPTO record (here, 1985–1994) cannot be scored cleanly: the look-behind window reaches into a period when the corpus is sparse, the reference distribution has high entropy, and any specific filing looks artificially novel. The second is symmetric: the most recent five years of filings (2021–2026) cannot be scored cleanly either, because the look-ahead window has not yet completed. Figure 1 shades both windows. We restrict all regressions to filings from 1995 onward, and we read 2020 as the latest cohort eligible to demonstrate any kind of post-filing fate.

4 Face validity

Figure 2 plots each filing in the prospective–retrospective KL plane. The horizontal axis (prospective KL) measures dissimilarity from filings in the previous five years (high ⇒ less like the past); the vertical axis (retrospective KL) measures dissimilarity from filings in the next five years (high ⇒ less like the future). Filings *below* the diagonal $KL_{\text{pros}} = KL_{\text{retr}}$ are innovators by our criterion — the past does a worse job of predicting them than the future does. Filings above the diagonal are tired. The labelled marks include expectation-confirming category-creators (OpenAI 2016, Claude 2023, Instagram 2011, White Claw Seltzer Works 2016, Rockstar 2002) and expectation-disconfirming late-cycle marks (Hard Mtn Dew 2021).

The bottled-water category illustrates a useful failure mode. The 2019 Liquid Death filing for canned water describes its goods as *still water; sparkling water* — the same lexicon as established

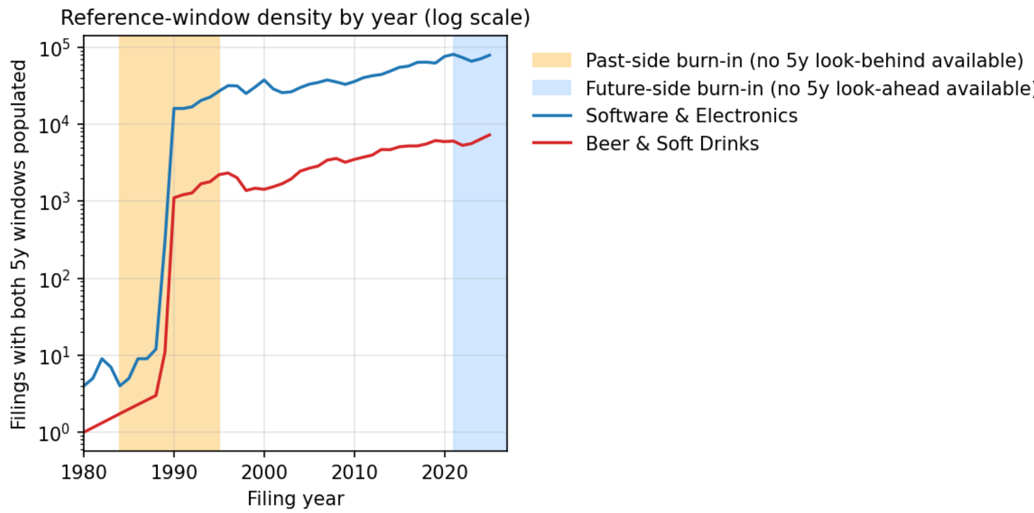

Figure 1: Filings per year with both 5-year reference windows populated. Amber: filings whose look-behind window reaches into pre-1985 sparse data. Blue: filings whose look-ahead window has not yet elapsed at the time of writing. Filings in either band are excluded from regressions.

bottled-water brands — and so lands near the diagonal. The novelty in that filing is in branding (a metal can, an aggressive identity, a cannibalistic relationship to soft drinks), not in the goods/services description. Comparable bottled-water filings (Saratoga, Smartwater, Essentia) cluster in the same region. The method is, by construction, blind to non-textual novelty.

5 Industry time series

Figures 3 and 4 show the per-year median prospective KL, retrospective KL, and ΔKL in each industry, with inter-quartile shading. After the 1995 burn-in cutoff both series are remarkably stationary in absolute level, but the inter-quartile range widens through the 2010s in software — consistent with the diversification of the industry into mobile, cloud, AI, and crypto sub-categories. Beverages show a similar but smaller widening.

6 Outcome 1: Trademark survival

We derive three binary outcomes per filing from the USPTO status code.

- **Completed registration:** the filing eventually reached a registered status (a non-null `registration_date` is in the record). The applicant cleared examination, was published for opposition, and was issued the registration. The lag from filing to registration is typically nine to twenty-four months and depends on examiner office actions, opposition activity, and whether the application was filed under intent-to-use. We treat any filing with a registration date as having "completed registration" regardless of the elapsed time.
- **Renewed 5y registration:** completed registration, was filed at least five years before the present, and was not cancelled. The first Section 8 maintenance filing is due between the fifth and sixth year of registration; failure to file it is the most common cause of early cancellation, so reaching the five-year mark is a meaningful viability test.

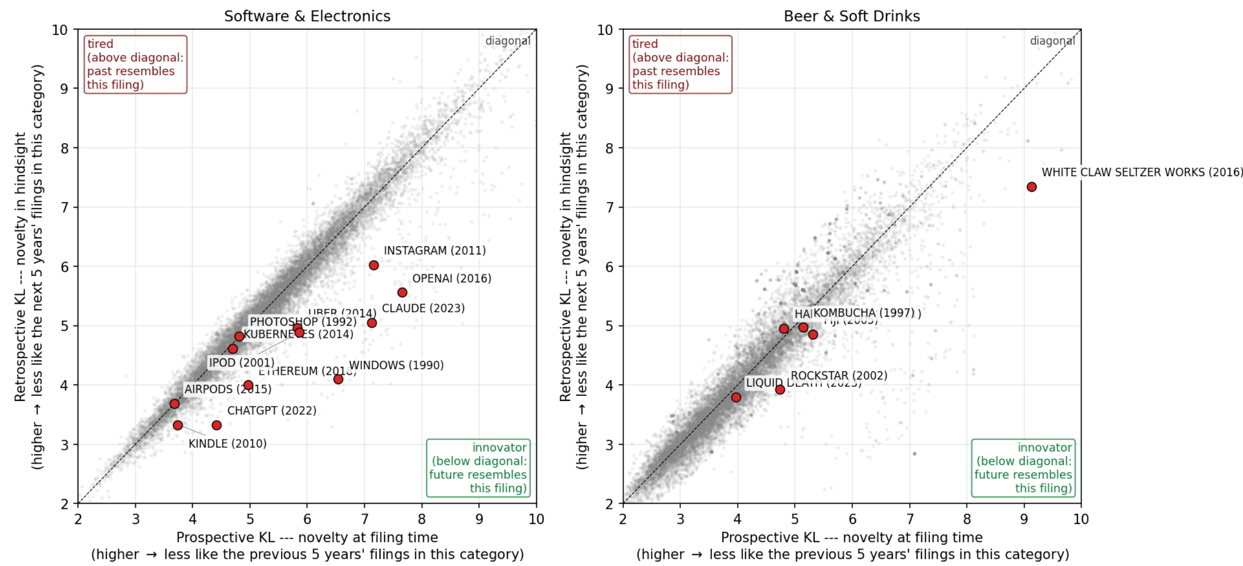

Figure 2: Per-filing prospective vs. retrospective KL surprise, zoomed to the central [2, 10] nat range where the labelled examples sit. Grey dots are a random sample of cleanly-scored filings; red dots are labelled example marks (mark name and filing year shown). **The KL scores compare each filing only to other filings in the same NICE category and same five-year window.** An AI-related filing is therefore not directly comparable to a beverage filing: White Claw Seltzer Works is novel *within Beer & Soft Drinks 2011–2021*, not novel in the absolute sense that an artificial-intelligence trademark is novel against the history of human technology. Below-diagonal: innovator (the future of the same category resembles this filing more than the past did). Above-diagonal: tired. Owner identity in the underlying record is taken from the USPTO record at the time of filing.

- **Renewed 10y registration:** completed registration, was filed at least ten years before the present, and was not cancelled. The first Section 9 renewal is due at year ten; subsequent renewals are due every ten years indefinitely.

We estimate generalized linear logit models of the form

$$\Pr(\text{outcome}_i = 1) = \text{logit}^{-1}\big(\beta_0 + \beta_1 \cdot z_i + \gamma_1 \log n_i + \gamma_2(t_i - \bar{t}) + \gamma_3(t_i - \bar{t})^2\big)$$

where z_i is a z-scored predictor (either ΔKL alone or KL^{pros} and KL^{retr} jointly), n_i is the count of distinct dictionary terms in filing i, and t_i is the filing year. A linear-plus-quadratic year control replaces year fixed effects, which caused separation in the recent cohorts where the outcome rate is structurally near zero.

Table 1: Logistic regressions of each survival outcome on ΔKL (novelty) and on the joint prospective + retrospective decomposition. Coefficients are in log-odds units per one standard deviation of the predictor; standard errors in parentheses.

Industry	Outcome (cohort)	N	β for ΔKL	β for pros (joint)	β for retr (joint)
Software & Electronics	Completed registration (all cohorts)	1,423,405	-0.163 (0.002)	-0.785 (0.009)	+0.895 (0.009)
Software & Electronics	Renewed 5y registration (filed $\leq$ 2020)	1,054,572	-0.039 (0.002)	-0.183 (0.011)	+0.228 (0.011)
Software & Electronics	Renewed 10y registration (filed $\leq$ 2015)	732,789	-0.020 (0.002)	-0.063 (0.012)	+0.123 (0.012)
Beer & Soft Drinks	Completed registration (all cohorts)	118,424	-0.121 (0.006)	-0.365 (0.022)	+0.455 (0.022)
Beer & Soft Drinks	Renewed 5y registration (filed $\leq$ 2020)	87,849	-0.097 (0.007)	-0.338 (0.026)	+0.344 (0.027)
Beer & Soft Drinks	Renewed 10y registration (filed $\leq$ 2015)	59,807	-0.044 (0.009)	-0.152 (0.032)	+0.130 (0.034)

The headline finding is the sign pattern. In Software & Electronics (n = 1,146,415), each one-standard-deviation increase in ΔKL — that is, the filing was more novel at the time than its eventual imitators were five years later — *lowers* the log-odds of completing registration by -0.042 (SE = 0.002) and lowers the log-odds of renewing the registration at five years by -0.043. Decomposed jointly, prospective surprise (novelty at filing time) penalizes every outcome, while retrospective surprise (the filing remained unusual even after five years of subsequent filings) reverses each sign. Read together, the disadvantage at the registration gate is concentrated among filings whose future does *not* resemble them — the unrewarded pioneers — while the long-run survival premium accrues to filings the future does come to resemble — the imitated innovators. Beer & Soft Drinks shows the same sign pattern at smaller magnitude.

Figures 5 and 6 display the per-bin registration-and-renewal rates against (left) prospective KL, (centre) retrospective KL, and (right) ΔKL. Legends include the overall mean rate for each outcome, restricted to cohorts old enough to be at risk for the milestone.

7 Outcome 2: Financial success

The trademark survival results in Section 6 establish that the innovation axis is meaningful at the per-filing level, but they say nothing about whether innovation pays the firm a financial return. To test that, we link USPTO owner names to SEC EDGAR firm names in two passes. The first pass aggressively normalises both sides (uppercase, strip punctuation, drop legal-entity suffixes such as `INC`, `LLC`, `CORP`, `HOLDINGS`, `LIMITED`, and so on) and accepts only exact matches on the normalised key. A second pass uses fuzzy token-sort matching (`rapidfuzz`, score cutoff 92) to capture the long tail of near-variants for USPTO owners with at least 10 filings. The combined crosswalk recovers 14,280 USPTO owners (9,370 exact + 4,910 fuzzy) covering 627,349 trademark filings (3.8% of the all-class corpus by filing count, but a much larger share by economic weight, because matched owners are by definition publicly-traded firms with material financial reporting). After joining to SEC firm-year financials and restricting to firm-years with non-degenerate gross margin and at least one trademark filing in the prior three years, the analysis panel contains 5,326 firm-years across 1,597 unique CIKs over fiscal years 2009–2024.

Filing volume is itself an innovation marker, but also a size marker. A firm filing many trademarks in a short window may be innovating across product lines — but it may also simply be a large firm with a large legal department. We separate the two by including the count of filings as a third predictor alongside the novelty score itself, and by controlling for log revenue throughout. Both effects turn out to be present and partially independent.

For each firm-year (c, y) we compute the trailing 3-year mean of the firm's filing-level scores: $\overline{\Delta KL}_{c,y}$, the prospective and retrospective components, the maximum within-window ΔKL (the firm's most-novel filing in the window), and the count of filings. We then estimate

$$y_{c,t} = \beta_0 + \beta_1 \cdot z(\text{novelty})_{c,t} + \gamma_1 \log \text{Revenue}_{c,t} + \gamma_2 \text{R\&D intensity}_{c,t} + \text{year controls} + \epsilon_{c,t}$$

with cluster-robust standard errors at the firm level. Outcomes are gross margin, net margin, and operating margin.

The pattern from Section 6 reproduces here at the firm-year level. A one-standard-deviation increase in 3-year trailing mean ΔKL — that is, a firm whose recent filings introduced wording that was unusual at the time but came to look ordinary in the years that followed — raises gross margin by +3.2 percentage points (SE 0.6). Decomposed jointly, prospective surprise (novel against the past) raises gross margin by +14.6 pp per standard deviation (SE 4.1), and retrospective surprise

Table 2: OLS regressions of profitability outcomes on trademark novelty, with cluster-robust standard errors at the firm level. Coefficients are in profitability-units per one standard deviation of the predictor (so a coefficient of +0.029 on gross margin means "+2.9 percentage points"). Each block is a separate specification.

Predictor (z-scored unless noted)	Coef.	SE	N
Outcome: Gross margin			
3y trailing mean novelty ΔKL	+0.032	0.006	5,319
3y trailing prospective surprise (novel vs. past)	+0.146	0.041	5,319
3y trailing retrospective surprise (novel vs. future)	-0.191	0.041	5,319
3y trailing max novelty	+0.033	0.006	5,319
3y trailing average filing count	+0.019	0.007	5,319
Outcome: Net margin			
3y trailing mean novelty ΔKL	-1.751	1.630	4,864
3y trailing prospective surprise (novel vs. past)	-8.884	9.824	4,864
3y trailing retrospective surprise (novel vs. future)	+10.966	10.809	4,864
3y trailing max novelty	-3.503	2.604	4,864
3y trailing average filing count	-2.335	1.871	4,864
Outcome: Operating margin			
3y trailing mean novelty ΔKL	-0.628	0.600	4,882
3y trailing prospective surprise (novel vs. past)	-4.376	3.782	4,882
3y trailing retrospective surprise (novel vs. future)	+4.406	4.026	4,882
3y trailing max novelty	-1.253	0.843	4,882
3y trailing average filing count	-0.536	0.378	4,882

(still novel from the future's perspective, i.e. a filing nobody imitated) lowers it by -19.1 pp per standard deviation (SE 4.1). The within-firm story is the same as the within-filing story: firms whose recent filings the future ends up resembling enjoy higher gross margins; firms whose recent filings nobody imitates do not. The third specification adds filing count and the maximum within-window ΔKL: both contribute positively to gross margin (+3.3 and +1.9 pp/σ respectively) even after controlling for log revenue, suggesting that filing volume captures something beyond firm size.

Variance. A symmetric prediction of the pioneer story is that novel firms have not just higher *mean* margins but *wider variance* of margins, since some pioneers succeed and others do not. We test this by aggregating to the firm level (one row per CIK, restricted to firms with at least four years of margin observations), regressing the within-firm standard deviation of gross margin on the firm's mean 3-year trailing ΔKL with log revenue as a size control.

Table 3: Variance regression: within-firm std. of gross margin on firm-mean novelty.

Predictor (z-scored unless noted)	Coef.	SE	N firms
3y trailing mean novelty ΔKL (firm avg)	+0.0176	0.0038	492
log revenue (firm avg)	-0.0141	0.0016	492

A one-standard-deviation increase in firm-mean ΔKL raises the within-firm gross-margin standard deviation by +0.018 in margin units (SE 0.004) on 492 firms. The pioneer-bet effect is real and quantitatively meaningful: high-novelty firms simultaneously earn higher mean margins (Table 2) and exhibit wider margin spread (Table 3).

8 Cross-industry comparison

We process all 45 NICE international classes through the same pipeline. Table 4 reports per-class summary statistics; Figure 8 ranks industries by mean filing-level ΔKL; Figure 9 reports the per-industry logit coefficient of ΔKL on each survival outcome; Table 5 lists the highest-ΔKL mark per industry, picked from the 2010–2020 cohort window where every class has dense filing volume.

Table 4: Per-industry summary statistics, restricted to filings from 1995–2020 with both 5-year reference windows populated.

Industry (NICE)	Filings	Clean	Owners	Years	ΔKL med.	ΔKL mean	5y reg. rate	Reg. rate
Tobacco & Smokers' Articles (034)	82,520	31,547	16,087	1995–2020	+0.15	+0.28	43%	48%
Trim, Lace & Notions (026)	84,034	33,395	23,121	1995–2020	+0.10	+0.16	57%	64%
Ropes, Tents & Textiles (raw) (022)	47,287	19,895	13,300	1995–2020	+0.08	+0.13	58%	68%
Yarns & Threads (023)	15,210	2,889	1,756	2006–2020	+0.10	+0.12	66%	72%
Household Utensils (021)	403,933	172,345	109,053	1995–2020	+0.06	+0.10	56%	64%
Carpets & Wall Hangings (027)	56,986	20,959	12,190	1995–2020	+0.07	+0.09	56%	64%
Hand Tools (008)	128,745	57,965	35,798	1995–2020	+0.06	+0.09	57%	68%
Furniture (020)	326,621	132,147	80,029	1995–2020	+0.04	+0.09	55%	64%
Firearms & Pyrotechnics (013)	37,403	11,844	6,016	1995–2020	+0.08	+0.09	53%	66%
Lighting, HVAC & Appliances (011)	333,507	160,109	90,464	1995–2020	+0.04	+0.07	59%	68%
Textiles & Bedding (024)	173,759	78,947	47,704	1995–2020	+0.05	+0.07	54%	62%
Wine & Spirits (033)	222,138	59,728	28,902	1995–2020	+0.03	+0.05	44%	52%
Jewelry & Watches (014)	234,908	91,669	59,847	1995–2020	+0.02	+0.04	54%	61%
Restaurants & Hotels (043)	285,610	122,371	74,815	2003–2020	+0.02	+0.03	50%	60%
Vehicles (012)	242,485	107,119	56,375	1995–2020	+0.00	+0.03	54%	65%
Leather & Luggage (018)	249,052	126,295	80,619	1995–2020	-0.01	+0.02	54%	61%
Medical Devices (010)	302,192	160,575	74,616	1995–2020	-0.02	+0.02	51%	61%
Cosmetics & Cleaning (003)	564,014	278,140	130,889	1995–2020	+0.02	+0.01	49%	56%
Medical & Personal Care (044)	312,270	167,607	97,110	2003–2020	-0.00	+0.01	53%	63%
Common Metals (006)	175,437	81,525	46,824	1995–2020	+0.01	+0.01	54%	70%
Toys, Games & Sporting Goods (028)	583,108	258,260	131,810	1995–2020	-0.02	+0.01	51%	58%
Legal & Personal Services (045)	203,476	114,693	74,406	2003–2020	-0.01	+0.00	54%	63%
Education & Entertainment (041)	1,344,519	815,077	471,515	1995–2020	-0.01	-0.00	52%	62%
Rubber & Insulators (017)	88,430	43,458	22,524	1995–2020	-0.00	-0.00	55%	71%
Lubricants & Fuels (004)	78,345	30,267	16,546	1995–2020	-0.02	-0.00	49%	61%
Materials Treatment (040)	145,590	83,213	52,597	1995–2020	-0.02	-0.01	54%	66%
Musical Instruments (015)	31,792	10,609	6,531	1995–2020	-0.00	-0.01	54%	68%
Software & Electronics (009)	1,807,636	1,054,572	521,819	1995–2020	-0.02	-0.01	50%	58%
Construction & Repair (037)	284,361	157,226	93,653	1995–2020	-0.00	-0.01	56%	69%
Machines & Power Tools (007)	286,930	135,331	69,563	1995–2020	-0.03	-0.01	57%	72%
Advertising & Retail (035)	1,418,345	899,682	528,027	1995–2020	-0.03	-0.02	52%	61%
Scientific & Tech Services (042)	1,148,416	635,128	352,645	1995–2020	+0.00	-0.02	52%	60%
Pharmaceuticals (005)	600,473	314,898	118,037	1995–2020	-0.02	-0.02	42%	49%
Coffee, Tea, Bakery & Spices (030)	428,854	173,569	89,747	1995–2020	-0.02	-0.02	47%	56%
Paper, Print & Office (016)	712,163	367,643	205,093	1995–2020	-0.02	-0.02	48%	58%
Chemicals & Industrial (001)	218,420	105,461	44,697	1995–2020	-0.02	-0.02	51%	66%
Clothing, Footwear & Headwear (025)	1,265,966	561,490	361,440	1995–2020	-0.05	-0.03	47%	53%
Beer & Soft Drinks (032)	206,304	87,849	43,714	1995–2020	-0.03	-0.03	41%	48%
Transport & Logistics (039)	187,535	105,370	59,203	1995–2020	-0.02	-0.03	53%	65%
Meat, Dairy & Prepared Foods (029)	253,936	114,185	55,196	1995–2020	-0.04	-0.04	46%	56%
Building Materials (019)	122,596	60,289	30,109	1995–2020	-0.02	-0.04	52%	66%
Agriculture & Live Animals (031)	136,656	52,623	26,983	1995–2020	-0.04	-0.04	48%	59%
Paints & Coatings (002)	64,892	28,286	14,228	1995–2020	-0.05	-0.04	53%	69%
Insurance & Financial (036)	602,901	354,316	174,543	1995–2020	-0.05	-0.05	51%	63%
Telecommunications (038)	230,267	153,225	78,422	1995–2020	-0.04	-0.06	47%	53%

Three patterns visible across the 45 classes. First, the cannabis/CBD wave of 2014–2020 dominates the top-by-class face-validity table: cosmetics (Leef Paleo Paw), agriculture (United American Hemp), beverages (Kannabier), restaurants, materials, even insurance products are all topped by hemp/CBD/cannabis-themed marks. The blockchain wave (2014–2018) sweeps Class 35

(OpenSea), Class 41 (Kingsland School of Blockchain), and Class 36 (Mosaic Token). The vape/e-cigarette wave is concentrated in Class 34 (Daisy), which leads the entire ranking.

Second, the most-and-least innovative industries are surprising. Tobacco & Smokers' Articles (+0.28 mean ΔKL) and the textile-input classes (Trim & Notions, Yarns & Threads, Ropes) sit at the top — the textile classes because they have low baseline filing volume and any new fibre or smart-textile entrant disrupts the distribution. The bottom of the list is dominated by mature regulated industries: Telecommunications, Insurance & Financial, Building Materials, and Agriculture. Neither end is what one would have predicted in advance.

Third, the per-industry survival betas (Figure 9) show that the "novelty-is-penalised-at-the-registration-gate" result reported in Section 6 is *not* universal. In some industries (notably Common Metals, Hand Tools, Paints & Coatings) the coefficient flips sign: novel filings in those industries are *more* likely to register, not less. A plausible reading is that in industries where category-creation is rare and meaningful, an examiner reading a novel goods/services description perceives a real new use rather than vague language. Software (and pharmaceuticals, chemicals) follow the opposite pattern: novel filings face slower examination, more office actions, and lower registration rates, consistent with examiners pushing back on speculative or boilerplate broadenings of established categories.

9 Discussion

The "arrows in the back" versus "better mousetrap" tension from Section 1 has both pieces empirically supported in this corpus. At the trademark gate, novelty is penalised: the most prospectively surprising filings (the ones whose wording was most unusual at the time) are less likely to clear examination, less likely to file Section 8 maintenance, less likely to renew at 10 years — the arrows. But conditional on survival, the same novelty axis is rewarded: marks the future imitates persist longer in the registrar's records, and at the firm-year level, public firms whose trademark portfolios sit in the prospectively-surprising half of the distribution earn higher gross margins than otherwise-comparable peers — the mousetrap. The split between the two comes from the retrospective component: filings the future does *not* come to resemble bear the cost of novelty without collecting the reward.

To make the underlying mechanics concrete, Table 6 reports a small set of marks together with the top dictionary tokens that drove their prospective and retrospective surprise. Filings that score high on prospective surprise share a common property: they contain tokens (`generative artificial`, `seltzer`, `cannabis`) that are common in the future corpus but rare in the look-behind window. Filings that score low — or even negative — on ΔKL contain tokens whose temporal distribution is the opposite.

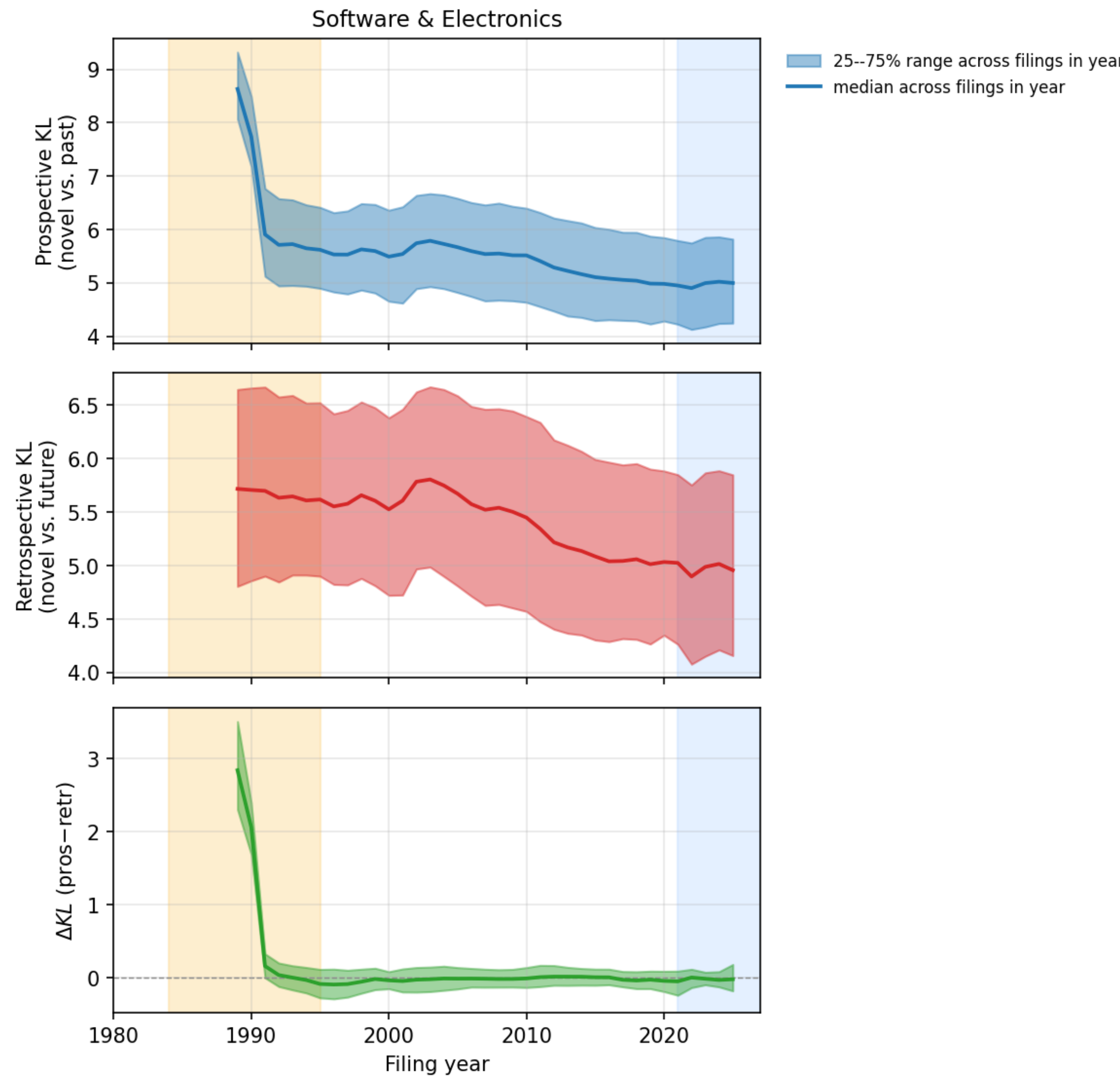

Figure 3: Software & Electronics: per-year median (line) and 25–75% range across all filings in that year (shaded band) of prospective KL, retrospective KL, and ΔKL. Amber band: filings without a complete 5-year look-behind window (artificially high prospective KL). Blue band: filings without a complete 5-year look-ahead window (artificially high retrospective KL). Excluded from all regressions.

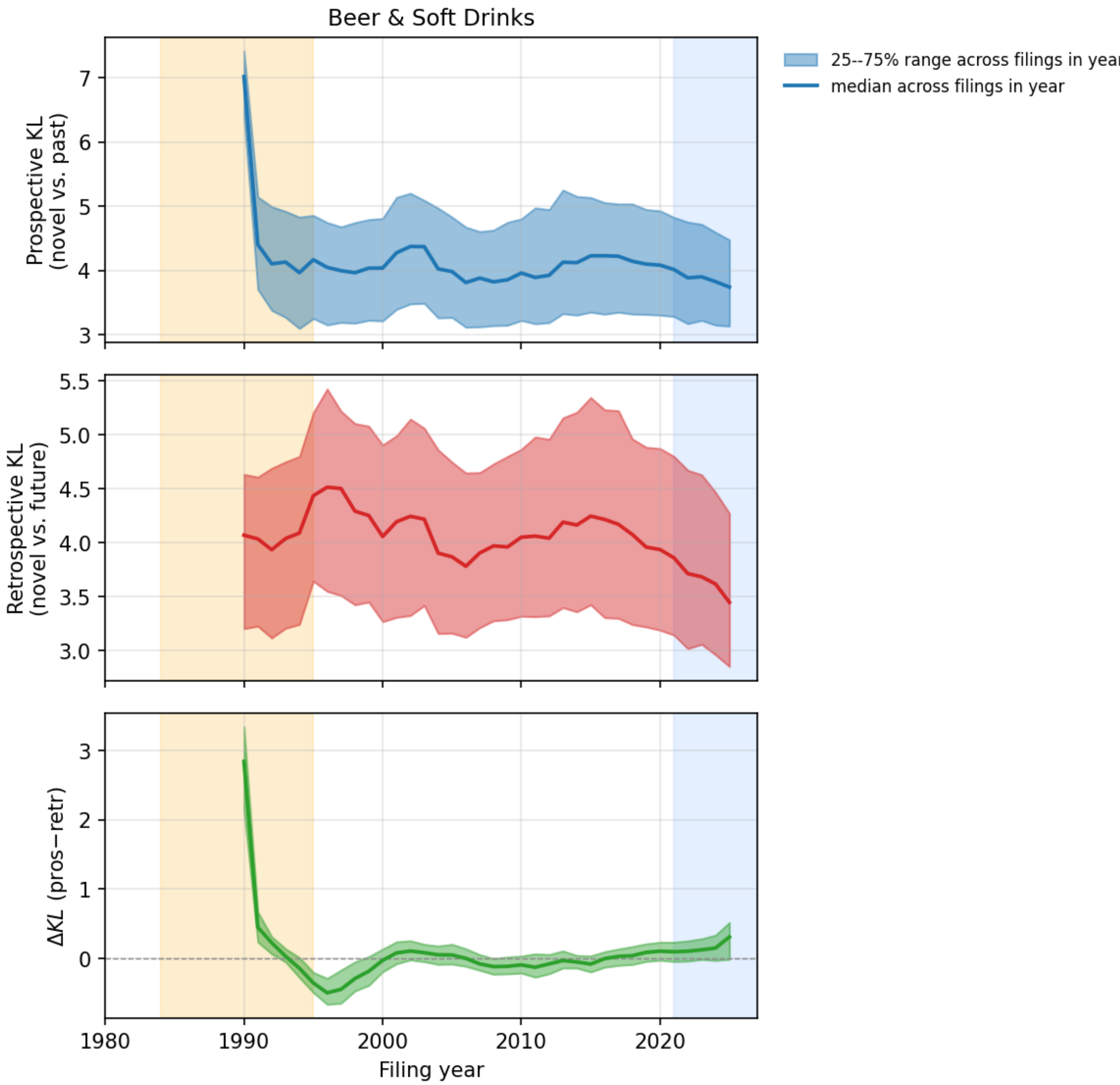

Figure 4: Beer & Soft Drinks: per-year median (line) and 25–75% range across filings in that year (shaded band) of prospective KL, retrospective KL, and ΔKL. Amber/blue: burn-in windows.

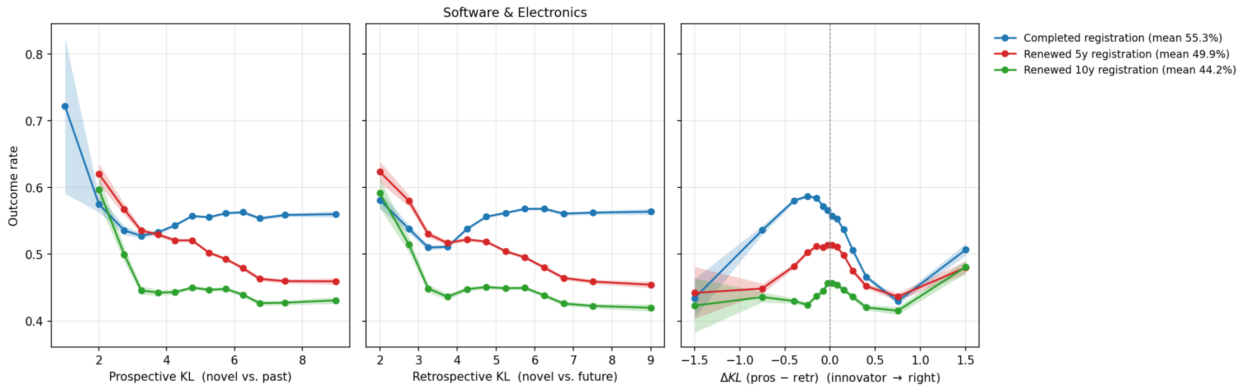

Figure 5: Software & Electronics: registration-and-renewal rates by KL bin. Shaded bands are 95% Wilson confidence intervals on the per-bin mean rate. Pattern across panels: high prospective KL (novelty against the past) penalises both registration and renewals; high retrospective KL (the wording remained unusual in hindsight) is neutral-to-favorable; the difference ΔKL inherits a downward slope.

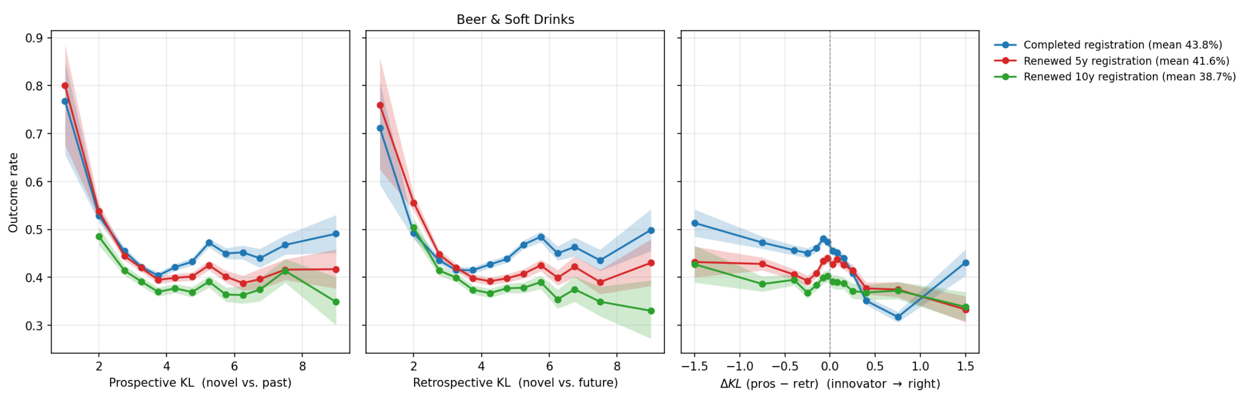

Figure 6: Beer & Soft Drinks: registration-and-renewal rates by KL bin. Shaded bands: 95% Wilson confidence intervals on the per-bin mean rate.

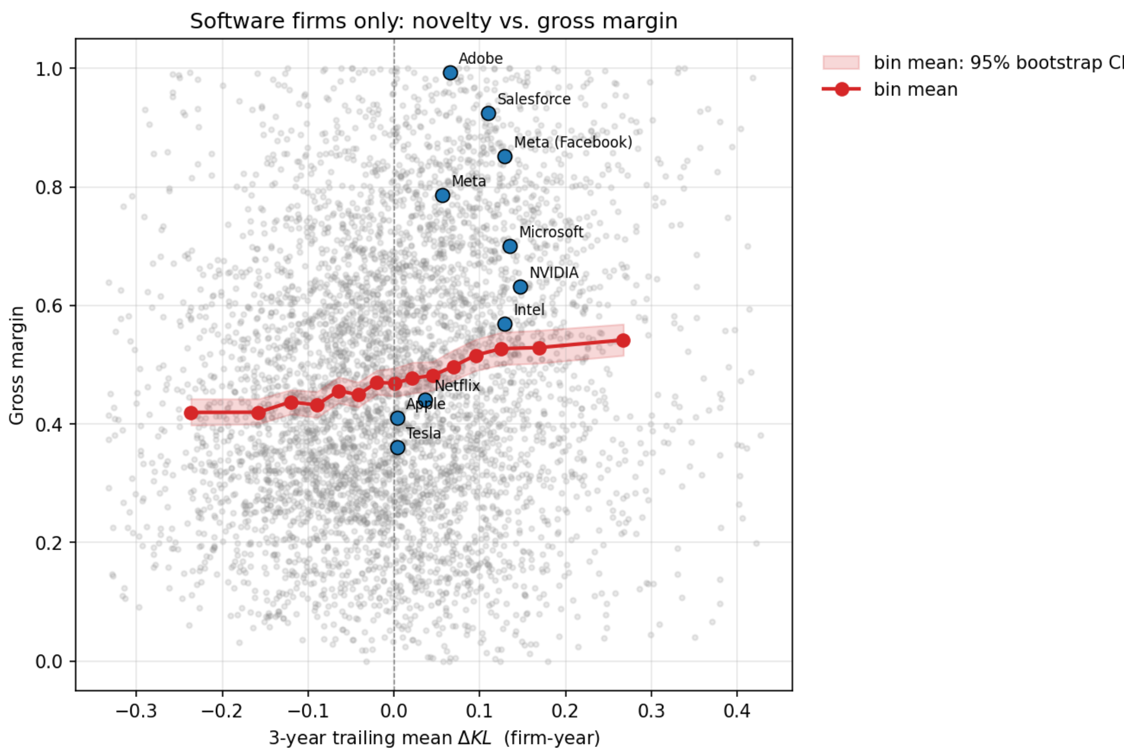

Figure 7: Software-only firm-year panel: gross margin vs. 3-year trailing mean ΔKL. Grey: the underlying 4,044 firm-year observations across 1,366 public software firms (1st–99th percentile of ΔKL, gross margin clamped to $[0, 1]$). Red line: bin means; red shaded band: 95% bootstrap confidence interval on each bin's mean gross margin. Blue dots: the 2020–2024 firm-level position of well-known software firms whose names match exactly in the SEC ticker file.

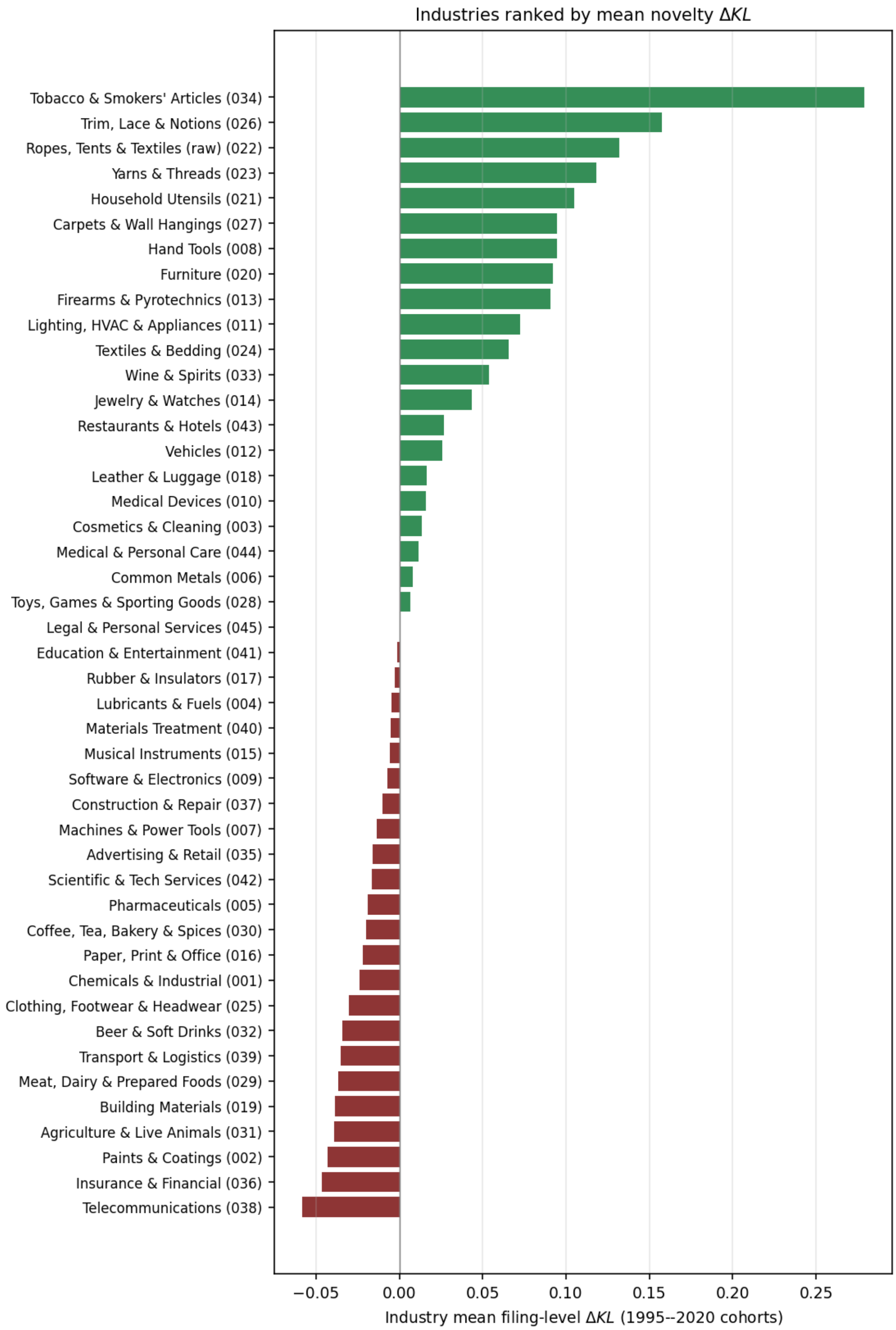

Figure 8: Industries ranked by mean filing-level ΔKL across 1995–2020 cohorts. Green: industries whose typical filing was prospectively novel and retrospectively obvious (the future imitated the past). Red: industries whose typical filing was the opposite. Tobacco & Smokers' Articles tops the list, driven by the 2014–2020 e-cigarette / vape vocabulary explosion. Telecommunications, Insurance & Financial, and Paints & Coatings sit at the bottom: their goods-and-services lexicon was already settled before 1995 and has been stable since.

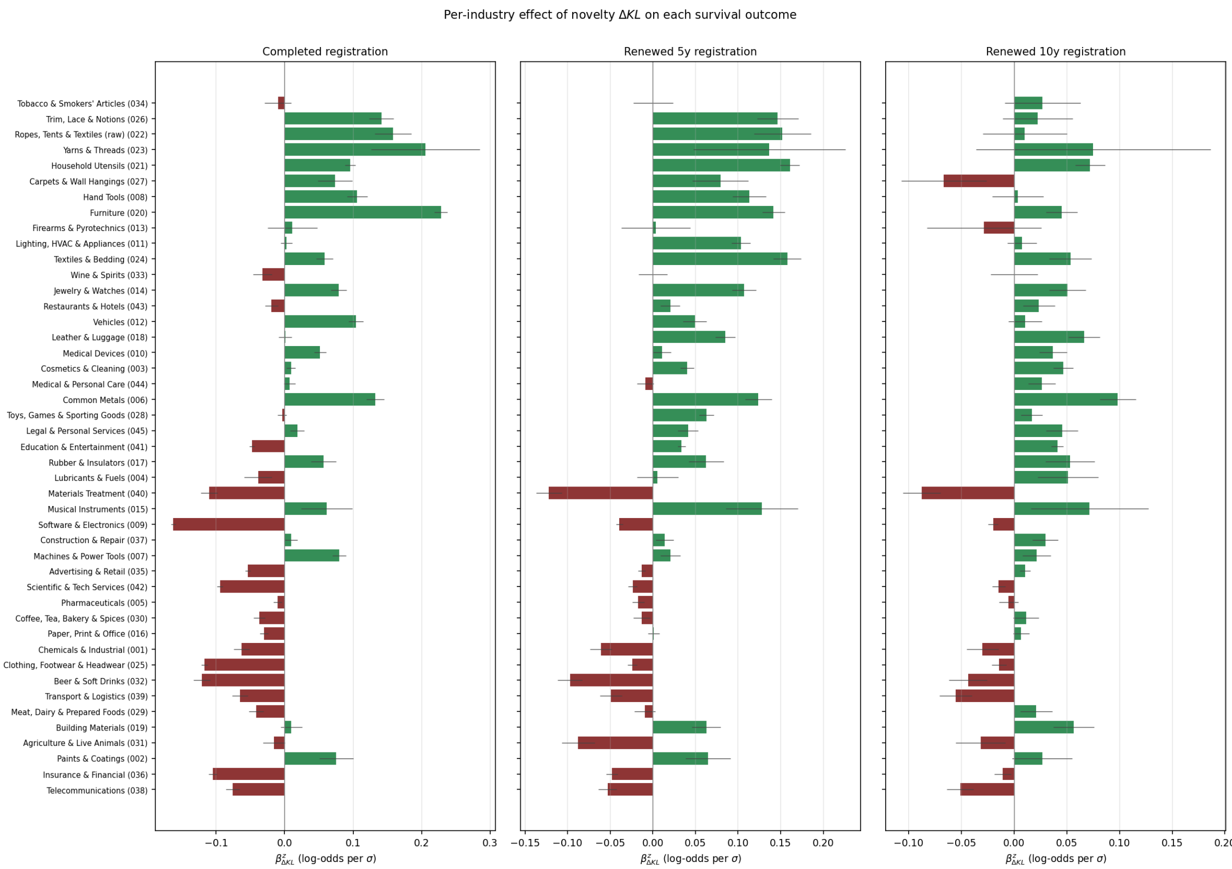

Figure 9: Per-industry effect of ΔKL on survival outcomes (logistic-regression z-score coefficient with 95% CI). Industries are ordered by mean ΔKL (the same order as Figure 8). **The sign of the innovation reward varies by industry.** Some industries (e.g. Common Metals, Hand Tools, Paints & Coatings) actually *reward* novel filings at the registration gate, presumably because in those industries a novel goods/services description can describe a real new use the examiner recognises as patentable activity. Most industries match the software pattern: novelty penalised at registration, neutral or favourable at renewal.

Table 5: Top-1 highest-ΔKL mark per industry, 2010–2020 cohort. The 2014–2020 cannabis/CBD/blockchain wave appears in nearly every consumer-goods class.

Industry (NICE)	Year	ΔKL	Mark	Owner at filing
Tobacco & Smokers' Articles (034)	2016	+4.51	DAISY	Green Tank Technologies Corp.
Agriculture & Live Animals (031)	2018	+4.20	UNITED AMERICAN HEMP	United American Hemp LLC
Cosmetics & Cleaning (003)	2018	+4.05	LEEF	PALEO PAW CORP.
Chemicals & Industrial (001)	2017	+4.03	REEFERTILIZER	Reefertilizer Inc.
Coffee, Tea, Bakery & Spices (030)	2012	+3.93	RED & WHITE	Forefront Enterprises, LLC
Household Utensils (021)	2017	+3.59	IREPELLER	10093939 Canada Corp.
Beer & Soft Drinks (032)	2018	+3.56	KANNABIER	Advanced Creative Intelligent
Wine & Spirits (033)	2014	+3.49	CHATEAU DE CAYX	Boutique du Château de Cayx
Ropes, Tents & Textiles (raw) (022)	2013	+3.33	SUNAMI	TENSILE SHADE PRODUCTS, LLC
Scientific & Tech Services (042)	2014	+3.30	ZENPOOL	Geniuses at Work Corporation
Toys, Games & Sporting Goods (028)	2012	+3.20	BETCLOUD	BALLY GAMING, INC.
Materials Treatment (040)	2018	+3.17	VERITAS — FARMS	271 Lake Davis Holdings LLC
Meat, Dairy & Prepared Foods (029)	2016	+3.15	PHIVIDA	PHIVIDA HOLDINGS INC.
Jewelry & Watches (014)	2013	+3.11	OFF THE CLOSET	Rahhal, Nour
Software & Electronics (009)	2018	+3.09		Unify Earth UEX
Furniture (020)	2016	+3.07	GO PET CLUB	Pel Wholesale, Inc.
Paper, Print & Office (016)	2020	+3.05	KAYLEEMAY LLC	Fall, Cayle M
Clothing, Footwear & Headwear (025)	2010	+3.03	CHARMANT FEMME	2BB UNLIMITED, INC.
Medical Devices (010)	2020	+3.03	ELASTIMASK	Bedford Industries, Inc.
Education & Entertainment (041)	2017	+3.01	ACADEMY	Kingsland School of Blockchain
Trim, Lace & Notions (026)	2015	+2.99	WOWQUEEN	XUE YONG BO
Advertising & Retail (035)	2018	+2.85	OPENSEA	Ozone Networks, Inc.
Lubricants & Fuels (004)	2017	+2.82		Mary Jane's Medicinals, LLC
Pharmaceuticals (005)	2017	+2.74	CANINES BEST DEFENSE CBD	Nobrega, Kathryn
Insurance & Financial (036)	2017	+2.71	MOSAIC TOKEN	MOSAIC RESEARCH LIMITED
Medical & Personal Care (044)	2018	+2.64	FARMING A HEALTHIER FUTURE	Southern Hemp, LLC
Carpets & Wall Hangings (027)	2016	+2.63	LOVEMAT	Wuhan Xuanyuanlong Science and
Transport & Logistics (039)	2017	+2.63	MARIJUANA EXPRESS	Marijuana Express LLC
Lighting, HVAC & Appliances (011)	2010	+2.42	KESSIL	DiCon Fiberoptics, Inc.
Vehicles (012)	2013	+2.34	PRECISION DRONE WWW.PRECISIONDRO	Precision Drone, LLC
Leather & Luggage (018)	2018	+2.31	NOMADZ	Ambit Global
Textiles & Bedding (024)	2017	+2.27	BESTDROP	Zhiyong Xie
Legal & Personal Services (045)	2011	+2.26	DIAMONDS IN THE RUFF	CINDY ROSS
Restaurants & Hotels (043)	2013	+2.20	GREEN FEET BREWING	Green Feet Brewing
Hand Tools (008)	2016	+2.02	THE NAZOR	Nazor, LLC
Firearms & Pyrotechnics (013)	2010	+1.94	ALPHA RAIL	TROY INDUSTRIES, INC.
Machines & Power Tools (007)	2017	+1.86	COUNTERMATE	IMPAK Corporation
Musical Instruments (015)	2012	+1.75	ERGODOCK	Hickey, Bruce
Common Metals (006)	2017	+1.70	UP YOUR CASTER	CASTER CONNECTION, LLC
Yarns & Threads (023)	2011	+1.69	BERROCO NANUK	Berroco, Inc.
Rubber & Insulators (017)	2014	+1.69	B3D	Brandon Bell
Construction & Repair (037)	2010	+1.52	WILKS MASONRY	Wilks Masonry Corporation
Building Materials (019)	2012	+1.51	FREE FORM YOGA FLOOR	Isabelle J. Barger
Telecommunications (038)	2017	+1.44	NCRYPT	NCHAIN HOLDINGS LTD.
Paints & Coatings (002)	2016	+1.32	TRIPLE BEST	Sike Wu

Table 6: Selected filings with per-token KL contributions to prospective and retrospective surprise. **Surprise is an interaction across tokens, not a property of any single word.** A filing's KL score is $\sum_v P(v)\log[P(v)/Q(v)]$, summed across every token v that appears in the filing's goods/services description. The token `software` is by itself common in Class 9, but its *combination* with rare neighbors — `generative artificial`, `conversational agents`, `cryptographic keys` — can give a filing high prospective surprise relative to the look-behind window's mix of tokens. Tokens shown below are the top five by per-token contribution $P(v)\log[P(v)/Q(v)]$ to each direction's KL; common tokens like `software` can appear in the prospective list when their multinomial weight in the filing is much larger than their weight in the look-behind reference.

Mark / Year	Owner at filing	KL_{pros}	KL_{retr}	ΔKL	Top tokens driving prospective surprise	Top tokens driving retrospective surprise
OPENAI (2016)	OpenAI, Inc.	7.65	5.56	+2.09	`software artificial`, `artificial intelligence`, `intelligence`, `artificial`, `software`	`software artificial`, `artificial intelligence`, `artificial`, `intelligence`, `software`
CLAUDE (2023)	Anthropic, PBC	7.12	5.06	+2.06	`model performing`, `document question-answering`, `tasks natural`, `generative text`, `performing generative`	`tasks`, `natural`, `model performing`, `question-answering simulating`, `document question-answering`
INSTAGRAM (2011)	INSTAGRAM, LLC	7.16	6.03	+1.13	`appearance enabling`, `modifying appearance`, `software modifying`, `enabling transmission`, `appearance`	`appearance enabling`, `modifying appearance`, `software modifying`, `enabling transmission`, `appearance`
PHOTOSHOP (1992)	ADOBE INC.	4.81	4.83	-0.01	`manipulating graphic`, `creating manipulating`, `images computer`, `graphic images`, `manipulating`	`manipulating graphic`, `creating manipulating`, `images computer`, `graphic images`, `programs creating`
KUBERNETES (2014)	LF PROJECTS, LLC	4.70	4.62	+0.08	`servers secure`, `secure private`, `private networks`, `networks`, `private`	`servers secure`, `secure private`, `private networks`, `networks`, `private`
WHITE CLAW SELTZER WORKS (2016)	MARK ANTHONY INTERNATIONAL S	9.12	7.35	+1.78	`sugar-based beer`, `sugar-based`, `seltzer`, `beer`	`sugar-based beer`, `sugar-based`, `seltzer`, `beer`
LIQUID DEATH (2023)	Supplying Demand, Inc.	3.97	3.80	+0.17	`drink mixes`, `water nutritional`, `mixes concentrates`, `water sparkling`, `concentrates`	`drink mixes`, `mixes concentrates`, `water nutritional`, `concentrates`, `flavor enhanced`
HARD MTN DEW (2021)	PEPSICO, INC.	4.81	4.96	-0.15	`flavored brewed`, `brewed`, `brewed malt-based`, `malt-based alcoholic`, `nature beer`	`flavored brewed`, `brewed`, `brewed malt`, `nature beer`, `malt-based alcoholic`
ROCKSTAR (2002)	ROCKSTAR ENERGY LLC	4.74	3.92	+0.81	`drinks`, `energy drinks`, `energy`	`drinks`, `energy drinks`, `energy`

Limitations

- **Goods/services text is one of several novelty channels.** Brand novelty (Liquid Death), design novelty (AirPods casing), and pricing novelty are invisible to this corpus. Estimates here should be read as innovation-in-business-description novelty.
- **The crosswalk is exact-match only.** A fuzzy-match second pass and manual review of the top-by-volume USPTO owners would plausibly double or triple the matched panel size and reduce survivorship bias toward firms that did not change their registered name.
- **Survivorship bias in the financial panel.** Firms appear in SEC FSDS only while public. Pre-IPO innovators (Anthropic, OpenAI, Stripe at most timepoints) and dead-but-once-public firms drop out. The financials reward we estimate is conditional on already being a public concern.
- **The reference distributions are smoothed multinomials, not topic models.** An LDA replication is the obvious artifact-check.

Next steps

The cross-industry universal sweep (all 45 NICE classes pooled) would pin down which industries are innovative by this measure and would allow new-class births (cannabis beverages 2018, generative AI tools 2022) to be scored against a cross-industry background. A fuzzy-match crosswalk would expand the financial panel from $\sim$ 1,400 firms toward the $\sim$5,000 ceiling of the SEC EDGAR universe. With either or both extensions in place, the natural follow-up is a *variance* regression: do high-ΔKL firms exhibit higher margin variance, the empirical fingerprint of pioneers either spectacularly succeeding or spectacularly failing?

A Per-class summary statistics

Industry (NICE)	Filings	Clean	Years	Owners	ΔKL med.	p5	p95
Software & Electronics (009)	1,807,636	1,515,300	1958–2025	754,869	-0.02	-0.36	+0.45
Beer & Soft Drinks (032)	206,304	125,512	1980–2025	62,180	+0.01	-0.53	+0.60

B Selected filings, full text

OPENAI (2016) Owner at filing: *OpenAI, Inc..* $KL_{\text{pros}} = 7.65$, $KL_{\text{retr}} = 5.56$, $\Delta KL = +2.09$.
Goods/services text: *Computer software for artificial intelligence*

CLAUDE (2023) Owner at filing: *Anthropic, PBC.* $KL_{\text{pros}} = 7.12$, $KL_{\text{retr}} = 5.06$, $\Delta KL = +2.06$.
Goods/services text: *Downloadable computer software in the nature of an artificial intelligence model for performing generative text AI tasks and natural language processing AI tasks and for writing content based on a theme, summarizing text, document question-answering and simulating natural conversation; downloadable software in the nature of a downloadable mobile application featuring an artificial intelligence model for performing generative text AI tasks and natural language processing AI tasks and for writing content based on a t...*

INSTAGRAM (2011) Owner at filing: *INSTAGRAM, LLC.* $KL_{\text{pros}} = 7.16$, $KL_{\text{retr}} = 6.03$, $\Delta KL = +1.13$.

Goods/services text: *Downloadable computer software for modifying the appearance and enabling transmission of photographs*

PHOTOSHOP (1992) Owner at filing: *ADOBE INC..* $KL_{\text{pros}} = 4.81$, $KL_{\text{retr}} = 4.83$, $\Delta KL = -0.01$.

Goods/services text: *computer programs for creating and manipulating graphic images on a computer [and manuals for use therewith, sold as a unit]*

KUBERNETES (2014) Owner at filing: *LF PROJECTS, LLC.* $KL_{\text{pros}} = 4.70$, $KL_{\text{retr}} = 4.62$, $\Delta KL = +0.08$.

Goods/services text: *Computer software for accessing, browsing, sharing, defining, maintaining, virtualizing and communicating information over computer networks, servers and secure private networks; computer software for use in creating, managing, controlling, changing, replicating, deploying, naming and linking information over computer networks, servers and secure private networks; computer software for assisting users in navigating through computer networks, servers and secure private networks; computer software for running web app...*

WHITE CLAW SELTZER WORKS (2016) Owner at filing: *MARK ANTHONY INTERNATIONAL SRL.* $KL_{\text{pros}} = 9.12$, $KL_{\text{retr}} = 7.35$, $\Delta KL = +1.78$.

Goods/services text: *Hard seltzer Brewed sugar-based beer*

LIQUID DEATH (2023) Owner at filing: *Supplying Demand, Inc..* $KL_{\text{pros}} = 3.97$, $KL_{\text{retr}} = 3.80$, $\Delta KL = +0.17$.

Goods/services text: *Powdered drink mixes for making flavored water, sports drinks, and energy drinks; powders and concentrates for making flavor enhanced water and sparkling water Powdered nutritional supplement drink mixes and concentrates*

HARD MTN DEW (2021) Owner at filing: *PEPSICO, INC..* $KL_{\text{pros}} = 4.81$, $KL_{\text{retr}} = 4.96$, $\Delta KL = -0.15$.

Goods/services text: *flavored brewed malt-based alcoholic beverages in the nature of beer Alcoholic flavored brewed malt beverages, except beers*

ROCKSTAR (2002) Owner at filing: *ROCKSTAR ENERGY LLC.* $KL_{\text{pros}} = 4.74$, $KL_{\text{retr}} = 3.92$, $\Delta KL = +0.81$.

Goods/services text: *Sports drinks, namely, energy drinks*